

Name: _____

UNIT 30 - MOLECULAR GENETICS

Topic 1: DNA Structure & Replication

DNA: The Hereditary Material

Summarize how each of the following experiments contributed to our understanding of DNA as the hereditary material.

1) GRIFFITH'S TRANSFORMATION EXPERIMENT

What they did	What it proved

2) HERSHEY-CHASE EXPERIMENT

What they did	What it proved

Name: _____

The Structure of DNA

DNA molecules form a double helix, with a sugar-phosphate backbone held together by complementary nitrogenous base pairs. The “rules” for each DNA molecule include...

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

THE DNA MOLECULE

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DNA Replication is Semiconservative

Complete the following table comparing the three different models of DNA replication:

	CONSERVATIVE	SEMICONSERVATIVE	DISPERSIVE
Parent DNA			
DNA after 1st replication			
DNA after 2nd replication			

- *Outline how Meselson & Stahl were able to conclude that the semiconservative model of DNA replication is correct.*

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DNA Replication

During S phase of interphase, a cell replicates its entire genome in order to produce a new cell with an identical set of DNA. Each strand of DNA serves as a template for the creation of a complementary strand (semiconservative replication).

STEP 1: INITIATION

1)

2)

3)

STEP 2: ELONGATION

4)

5)

6)

7)

STEP 3: TERMINATION

8)

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9)

RE-CAP: ENZYMES INVOLVED IN DNA REPLICATION

Enzyme	Function
Helicase	
Primase	
DNA polymerase I	
DNA polymerase III	
DNA ligase	